

MASK WHAT MATTERS: CONTROLLABLE TEXT-GUIDED MASKING FOR SELF-SUPERVISED MEDICAL IMAGE ANALYSIS

Anonymous authors
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ABSTRACT

The paucity of annotated data that characterizes specialized domains such as medical imaging constitutes a formidable impediment to the training of robust vision models. Although self-supervised masked image modeling (MIM) furnishes a compelling avenue for mitigating this constraint, prevailing formulations predominantly hinge upon indiscriminate high-ratio masking, an expedient that engenders both computational inefficiency and impoverished semantic alignment. Region-aware variants, for their part, are ordinarily contingent upon reconstruction heuristics or supervisory signals, a dependence that curtails their adaptability across heterogeneous tasks and modalities. We advance Mask What Matters, a controllable text-guided masking framework devised for self-supervised medical image analysis. By harnessing vision-language models to effect prompt-based region localization, the proposed method adaptively confers differentiated masking so as to accentuate diagnostically salient regions while attenuating the redundancy inherent to background areas. This controllable formulation, in turn, promotes tighter semantic alignment, more expressive representation learning, and enhanced cross-task generalizability. A comprehensive evaluation spanning multiple medical imaging modalities—encompassing brain MRI, chest CT, and lung X-ray—demonstrates that Mask What Matters consistently surpasses established MIM methods (e.g., SparK), attaining improvements of as much as +3.1 percentage points in classification accuracy, +1.3 in box average precision (BoxAP), and +1.1 in mask average precision (MaskAP) for detection. Notably, these gains are realized under markedly lower aggregate masking ratios (e.g., 40% as opposed to 70%). Collectively, this work substantiates the proposition that controllable, text-driven masking can engender semantically aligned self-supervised learning, thereby advancing the development of robust vision models for medical image analysis.

1 INTRODUCTION

Image classification, segmentation, and object detection constitute the canonical pillars of computer vision, and machine learning—deep learning most conspicuously among its variants—has, over the preceding years, emerged as the core technology propelling progress throughout the field. Yet the efficacy of high-performing deep learning models is contingent, almost without exception, upon the availability of large-scale annotated datasets. Within specialized domains such as medical image processing, where high-quality annotations are at once scarce and prohibitively costly to procure, this dependence erects a substantial barrier to the broad assimilation of deep learning methods.

Self-supervised learning (SSL) furnishes a compelling remedy by exploiting unlabeled data for pre-training, thereby enabling models to acquire effective feature representations in the absence of manual annotation [Jaiswal et al. (2020); Liu et al. (2021); Krishnan et al. (2022); Huang et al. (2023)]. Among the several SSL paradigms, MIM has come to prominence: by randomly occluding portions of an image and compelling the model to reconstruct the original content, MIM induces the capture of structural and semantic regularities from the visible context. Its efficacy across a spectrum of downstream tasks has been amply demonstrated, and it has been successfully transposed to medical image analysis [He et al. (2022); Gupta et al. (2025); Xiao et al. (2023)].

Prevailing MIM formulations, however—exemplified by masked autoencoders (MAE) [He et al., 2022] and SparK [Tian et al., 2023]—characteristically resort to random masking strategies that stand unaligned with the regions of interest germane to downstream tasks. Such task-agnostic masking engenders two principal limitations: (1) the learned representations are liable to semantic misalignment with the downstream objectives, thereby diminishing transfer performance; (2) in an effort to raise the probability of occluding informative areas, extant methods frequently adopt inordinately high masking ratios (e.g., 75% in MAE), a practice that compounds reconstruction difficulty while exacting large-scale data and considerable computational resources. Natural language, by contrast, exhibits a compact structure and elevated semantic density that permit effective learning even under modest masking ratios [Devlin et al., 2019]. The reliance on high masking ratios that pervades image-based MIM stems largely from intrinsic information redundancy together with the absence of semantic-aware masking, a conjunction that impedes the model’s capacity to concentrate upon genuinely critical regions.

This predicament is thrown into particularly sharp relief in medical imaging. Medical images are marked by pronounced semantic sparsity, whereby diagnostically pertinent information is frequently sequestered within small localized regions (e.g., lesions or organs), while a preponderant share of the image comprises semantically redundant background. Compounding this, medical imaging subsumes a diversity of modalities—MRI, CT, and X-ray among them—across which the shape, extent, and spatial disposition of task-relevant areas vary substantially from task to task. These properties, taken together, impose exacting demands upon the efficiency and generalizability of any masking strategy.

To confront these challenges, we introduce a controllable, text-guided masking framework, hereafter designated **Mask-What-Matters (MWM)**. Given a user-specified, open-vocabulary description or phrase, a pretrained vision–language model supplies zero-shot localization cues that we subsequently transmute into robust regions of interest (ROIs). MWM thereupon administers region-specific, prompt-conditioned masking—apportioning higher ratios to semantically consequential areas (e.g., lesions/organs) and lower ratios to background—and in so doing injects downstream semantics directly into the pretraining signal, dispensing entirely with per-image reports or labels.

To summarize, the principal contributions of this work are as follows:

- The first controllable text-guided masking framework conceived for medical imaging. By integrating vision–language models (BiomedCLIP) with segmentation refinement (SAM), MWM localizes task-relevant regions from open-vocabulary prompts and thereupon administers differentiated ROI-versus-background masking, thereby redressing the semantic misalignment endemic to random masking.
- An annotation-free and backbone-agnostic design. MWM obviates any per-image report or label whatsoever and admits seamless incorporation into both ViT- and ConvNet-based MIM pipelines under a single unified protocol.
- Consistent gains across modalities and tasks. On brain MRI, chest CT, and X-ray datasets alike, MWM eclipses state-of-the-art MIM baselines (e.g., SparK)—improving classification, detection, and segmentation performance in concert—and thereby attests to the promise of text-driven generative pretraining in medical imaging.

2 RELATED WORK

2.1 MASKED IMAGE MODELING

Masked modeling had its genesis within the natural language processing domain, with the bidirectional encoder representations from transformers (BERT) model [Devlin et al., 2019], which acquires contextual representations by occluding tokens and predicting them over the course of pretraining. Drawing inspiration from this conception, [He et al., 2022] transposed MIM to computer vision, wherein portions of an image are masked during pretraining and the model is charged with reconstructing the missing regions. This paradigm has evinced robust generalization across both the natural and medical image domains [Gupta et al., 2025; Liu et al., 2021; Huang et al., 2023]. Representative instantiations include MAE [He et al., 2022], MedMAE [Gupta et al., 2025; Xiao et al., 2023], and SparK [Tian et al., 2023]. Of these, MAE marshals high-ratio random masking to impel ViTs toward high-level semantic representations; MedMAE extends this conception to medical

images by way of domain-specific adaptations; and SparK recasts the strategy for convolutional backbones, capitalizing on their intrinsic multi-scale features to augment the efficacy of pretraining.

Notwithstanding these advances, conventional MIM approaches characteristically embrace random or structure-agnostic masking strategies that remain oblivious to the semantic exigencies of downstream tasks. Such non-selective masking is apt to induce the model to lavish undue emphasis upon irrelevant regions during pretraining, thereby eroding transfer effectiveness. In an effort to remedy this deficiency, recent work has investigated region-aware masking strategies directed at improving semantic alignment, along two principal axes:

The first line of inquiry advances loss-driven adaptive masking, wherein methods such as hard patches mining (HPM) [Wang et al. (2023a)] and its extension AnatoMask [Li et al. (2024)] dynamically scrutinize reconstruction errors across regions and confer higher masking probabilities upon areas exhibiting larger losses, with the aim of prioritizing semantically valuable structures. This approach, however, can misdirect the model toward regions that are arduous to reconstruct yet task-irrelevant. In brain magnetic resonance imaging (MRI), for instance, intricate structures such as the spinal cord or the eyeballs may command superfluous attention notwithstanding their irrelevance to downstream objectives such as tumor detection.

A second line of inquiry enlists external perceptual modules to steer masking. FocusMAE [Basu et al. (2024)], for example, exploits pretrained Region Proposal Networks (RPNs) to identify high-information regions for selective masking. Efficacious though it proves on certain tasks, this strategy presupposes the supervised training of RPNs upon domain-specific annotated data, a requirement that constrains its cross-task generalization.

These limitations, in aggregate, accentuate the need for a masking mechanism at once more flexible, more semantically aligned, and more generalizable. Natural language, as a rich and interpretable wellspring of supervision, presents an auspicious avenue for steering masked modeling in a manner that is task-aware yet altogether annotation-free.

2.2 VISION-LANGUAGE PRE-TRAINED MODELS

The rapid maturation of vision-language alignment models in recent years has rendered it increasingly tractable to perceive and delineate task-relevant regions with precision under zero-shot and open-world regimes [Radford et al. (2021)]. Canonical approaches such as the Contrastive Language-Image Pretraining (CLIP) family instate a unified embedding space for images and texts by way of large-scale joint pretraining on image-text pairs, and have accordingly been embraced across zero-shot and open-set visual tasks. Within the medical imaging domain, extensions such as Biomed-CLIP [Zhang et al. (2023)] further fortify cross-modal understanding, attaining robust performance across a diversity of downstream applications. Yet although vision-language pretrained models have been assimilated widely into an assortment of supervised learning tasks, the preponderance of extant studies remains fixated upon downstream applications such as object detection and semantic segmentation [Zhong et al. (2022), Aleem et al. (2024), Koleilat et al. (2024)], leaving unexplored the question of how their open-vocabulary and zero-shot faculties might be turned to the benefit of masked image modeling.

In summation, although recent region-aware MIM approaches—among them dynamic masking predicated on reconstruction loss and masking steered by external perception modules—have to some degree sharpened semantic focus, they remain encumbered by several limitations: a reliance on supplementary supervision, a misalignment between masked regions and task semantics, and a want of flexibility. Concurrently, notwithstanding the compelling performance of vision-language models on downstream tasks, scant effort has been devoted to enlisting natural language prompts within the MIM pretraining phase for the purpose of task-aware masking guidance. These observations, taken jointly, adumbrate an auspicious direction—the incorporation of vision-language semantic guidance so as to enable efficient and generalizable masked image modeling.

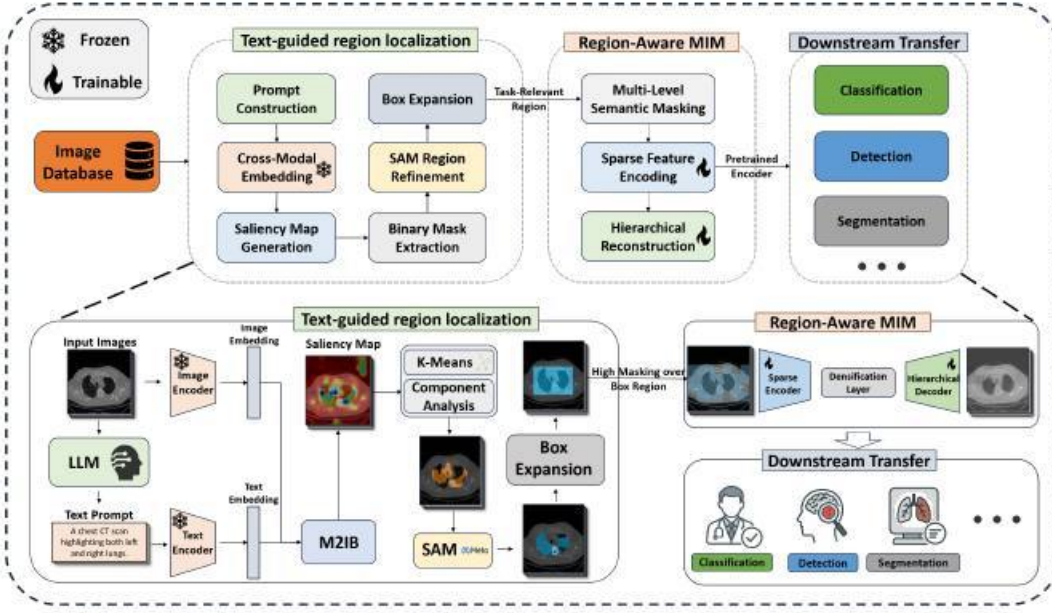


Figure 1: Overview of the MWM framework. The upper panel delineates the three-stage pipeline: (1) text-guided region localization, (2) region-aware masked image modeling, and (3) downstream transfer. The lower panel affords a magnified view, illustrating the manner in which prompts orchestrate region localization and masking.

3 METHODOLOGY

3.1 OVERALL FRAMEWORK

As illustrated in Figure 1, **Mask What Matters (MWM)** is constituted of two core stages: (1) text-guided region localization and (2) region-aware masked image modeling. The former stage enlists a frozen vision-language model to discern task-relevant regions from an open-vocabulary prompt, whereas the latter confers differentiated masking ratios across regions so as to steer the reconstruction process.

3.2 TEXT-GUIDED REGION LOCALIZATION

MWM localizes task-relevant regions by means of a multi-stage pipeline. It commences with LLM-based prompt generation and subsequently deploys BiomedCLIP to extract cross-modal embeddings. A saliency map is thereupon engendered via the Multi-Modal Information Bottleneck (M2IB), binarized with K-Means, and refined through the Segment Anything Model (SAM). The refined mask is at last transmuted into a bounding box endowed with a controllable expansion margin, which thereafter serves as a robust spatial guide for the ensuing masked modeling stage.

Prompt Generation. In medical image analysis, the crafting of textual prompts assumes a decisive role in dictating the efficacy of region localization. To amplify semantic guidance, large language models (LLMs) are interrogated with structured prompts such as “Describe the typical visual characteristics of a [category] in a [modality] image”. Beyond such full-sentence prompts, terse category phrases are likewise enlisted experimentally as alternative inputs. A detailed juxtaposition of these prompt styles is furnished in the experimental section.

Cross-Modal Embedding. Once the prompt has been generated, a BiomedCLIP model [Zhang et al. \(2023\)](#) is enlisted to encode the medical image together with its prompt into a shared semantic space. Given an image I and a prompt T , the embeddings are computed as:

$$Z_{\text{img}} = \Phi_{\text{img}}(I), \quad Z_{\text{text}} = \Phi_{\text{text}}(T) \quad (1)$$

where Φ_{img} and Φ_{text} denote the vision and language encoders, respectively. These cross-modal embeddings constitute the semantic bedrock upon which saliency estimation rests in the subsequent stages.

Saliency Map Generation. A cross-modal saliency estimation module predicated on the Multi-modal Information Bottleneck (M2IB) [Wang et al. (2023b)] is marshaled to localize task-relevant visual regions, aligning image and text embeddings while concurrently suppressing modality-specific redundancy. Formally, the saliency map $\lambda_S \in [0, 1]^{H \times W}$ apports an importance weight to each spatial location commensurate with its semantic relevance to the prompt. M2IB accomplishes this by optimizing the following objective:

$$\mathcal{L}_{\text{M2IB}} = \text{MI}(Z_{\text{img}}, Z_{\text{text}}) - \gamma \cdot \text{MI}(Z_{\text{img}}, I) \quad (2)$$

where Z_{img} and Z_{text} are the image and text embeddings encoded by BiomedCLIP, respectively, and γ arbitrates the trade-off between the preservation of cross-modal relevance and the excision of task-irrelevant visual information.

Binary Mask Extraction. Having obtained the saliency map, we binarize it by means of unsupervised K-Means clustering [Lloyd (1982)] in order to localize the region designated by the text prompt, thereby engendering a preliminary foreground mask. Concretely, the map is partitioned into two pixel groups, and the cluster exhibiting the higher saliency values is adjudged foreground. To refine the mask further and to suppress spurious activations, connected component analysis is conducted upon the binary map, whereby only the largest foreground regions are retained. This step serves to filter out noisy peripheral areas and to concentrate attention upon the core semantic structures.

Region Refinement. To heighten spatial fidelity, SAM is incorporated as a refinement module [Kirillov et al. (2023)]. For each selected connected region $c_i \in \mathcal{C}^*$, its minimal enclosing bounding box is computed so as to serve as the visual prompt supplied to SAM.

Given the original image I together with the set of bounding boxes, SAM predicts a refined segmentation mask:

$$M_{\text{SAM}} = \text{SAM}(I, \text{Box}(c_i)) \quad (3)$$

The resultant mask M_{SAM} furnishes a more faithful rendering of the spatial structure.

Box-Based Region Expansion. The SAM mask procured at the preceding stage harbors boundary noise liable to misdirect the model’s focus during representation learning. To bolster the robustness of region guidance, the refined SAM mask is accordingly transmuted into a bounding box endowed with a controllable expansion margin, which thereupon serves as the definitive region of interest (ROI) for downstream masked modeling.

3.3 REGION-AWARE MASKED IMAGE MODELING

Given the region of interest engendered by the localization module in Section 3.2, MWM prosecutes masked image modeling by way of a three-stage pipeline: (1) a multi-level masking strategy that confers differentiated masking ratios in accordance with region importance, (2) a sparse encoder that processes the unmasked patches alone, and (3) a hierarchical decoder that reconstructs the full image from sparse multi-scale features.

Multi-Level Semantic Masking. MWM embraces a multi-level masking strategy that capitalizes upon the semantic regions discerned by the text-driven localization process. Specifically, patches corresponding to regions foregrounded by the textual prompts—such as those associated with tumors or organs—are masked at a high ratio, so as to impel the model to concentrate upon reconstructing task-relevant semantic features. Background regions, by contrast, are masked but lightly or else left unmasked altogether, thereby curtailing redundant computation. A visual juxtaposition of the masking patterns is presented in Figure 2.

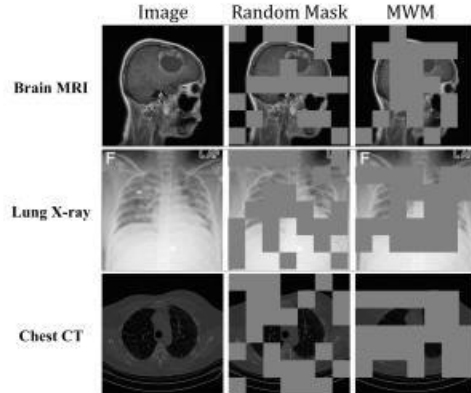


Figure 2: A comparison of masking strategies, wherein gray blocks demarcate the masked regions.

Sparse Feature Encoding. The visible patches are aggregated into a sparse image and encoded in accordance with the SparK design [Tian et al. \(2023\)](#):

$$\begin{aligned} X_{\text{sparse}} &= \{x_i \mid m_i = 1\}, \\ F &= \Phi_{\text{sparse}}(X_{\text{sparse}}) \end{aligned} \quad (4)$$

where x_i denotes the i -th patch, $m_i \in \{0, 1\}$ signals visibility, Φ_{sparse} is the encoder, and $F = \{F_1, F_2, F_3, F_4\}$ are the multi-scale features. Although we instantiate MWM with convolutional backbones, the identical interface extends to ViTs by way of token indexing.

Hierarchical Reconstruction. A lightweight UNet-style decoder reconstructs the full image in a top-down fashion. At each scale l , the masked locations are populated with a learned embedding M_l :

$$F'_l(i) = \begin{cases} F_l(i), & \text{if patch } i \text{ is visible,} \\ M_l, & \text{if patch } i \text{ is masked.} \end{cases} \quad (5)$$

The densified map F'_l is projected through a scale-specific layer ϕ_l so as to obtain D_l , whereupon it is upsampled and fused with the features at the next lower level:

$$D_{l-1} = B_{l-1}(D_l) + \phi_{l-1}(F'_{l-1}). \quad (6)$$

Finally, the model is trained by minimizing the mean-squared error computed over the masked patches:

$$\mathcal{L}_{\text{recon}} = \frac{1}{|\mathcal{I}_{\text{mask}}|} \sum_{i \in \mathcal{I}_{\text{mask}}} \|\hat{x}_i - x_i\|^2, \quad (7)$$

where $\mathcal{I}_{\text{mask}}$ denotes the set of masked patch indices, x_i the original patch, and $\hat{x}_i$ its reconstruction.

4 EXPERIMENTS

4.1 DATASETS

Self-Supervised Pretraining Datasets. Three datasets, spanning disparate imaging modalities, are enlisted for self-supervised pretraining. (1) *Brain MRI* comprises approximately 17,000 images drawn from Brain Tumor MRI [Cheng et al. \(2015\)](#), BRISC [Fateh et al. \(2025\)](#), and BraTS 2018 [Menze et al. \(2014\)](#), encompassing a variety of tumor types and anatomical structures. (2) *Lung X-ray* consists of some 39,000 chest radiographs sourced from COVID-QU-Ex [Rahman et al. \(2021\)](#); [Chowdhury et al. \(2020\)](#), spanning COVID-19, non-COVID pneumonia, and normal cases. (3) *Chest CT* is composed of 12,000 slices extracted from a lung disease dataset [Konya \(2020\)](#), together with segmentation masks of fibrotic lungs derived from 107 patients.

Downstream Fine-Tuning Datasets. Three datasets are marshaled for downstream evaluation across classification and detection tasks. (1) *Brain Tumor MRI* [Feltrin \(2023\)](#) comprises 4,479 MRI

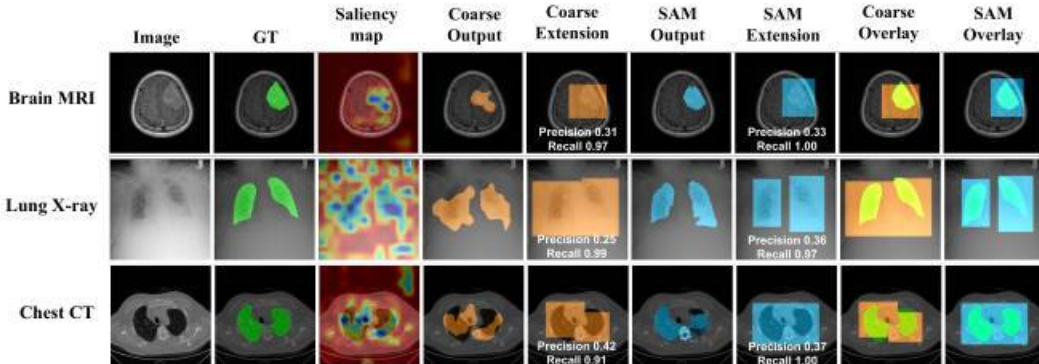


Figure 3: Representative outcomes of text-guided region localization.

Method	Brain MRI		Chest CT		Lung X-ray	
	Prec.	Rec.	Prec.	Rec.	Prec.	Rec.
<i>Prompt Type</i>						
Phrase	0.22	0.86	0.21	0.99	0.42	0.82
Descrip. Sent.	0.30	0.94	0.37	0.88	0.44	0.66
<i>Impact of SAM</i>						
Coarse Ext.	0.22	0.95	0.36	0.87	0.37	0.84
+ SAM Ext.	0.30	0.94	0.37	0.88	0.42	0.82

Table 1: Localization performance (Occlusion Precision/Recall) across prompt types and refinement strategies. *Coarse Ext.* designates a bounding-box extension of the coarse region, whereas + *SAM Ext.* denotes its further refinement by means of SAM.

images spanning 44 tumor types, among them astrocytoma, glioblastoma, meningioma, and ependymoma. (2) *Pediatric Lung X-ray* Kermany et al. (2018) consists of 5,863 pediatric chest radiographs, each categorized as either normal or pneumonia. (3) *Chest CT Cancer* Hany (2020) comprises CT images distributed across four classes: adenocarcinoma, large cell carcinoma, squamous cell carcinoma, and normal tissue.

4.2 EXPERIMENTAL SETUP

All self-supervised methods are furnished with backbones of comparable model scale, the convolution-based methods adopting ResNet-50 He et al. (2016) and the transformer-based methods employing ViT-S Dosovitskiy et al. (2020). The input resolution is fixed at 224×224, and the models are initialized with ImageNet-pretrained weights. Pretraining proceeds from an initial learning rate of 2×10^{-4} , while downstream fine-tuning preserves the identical encoder and input resolution. All experiments are conducted on NVIDIA Tesla A100 cards endowed with 80 GB VRAM, under Python 3.9, Ubuntu 22.04.3, and PyTorch 1.10.0. Unless otherwise stipulated, non-target variables (e.g., prompt type or SAM usage) are held fixed at their empirically optimal settings throughout evaluation, so as to guarantee a controlled comparison.

4.3 EVALUATION OF TEXT-GUIDED LOCALIZATION

We first appraise the efficacy of the text-guided localization module across three imaging modalities—brain MRI, lung X-ray, and chest CT—and thereafter scrutinize the bearing of prompt type and SAM refinement upon localization performance. Concretely, we juxtapose the predicted bounding boxes against expert-annotated segmentation masks, reporting occlusion precision ($|R_p \cap R_{gt}|/|R_p|$) and occlusion recall ($|R_p \cap R_{gt}|/|R_{gt}|$), wherein R_p denotes the predicted region and R_{gt} the ground-truth annotation.

As reported in Table 1 our text-guided localization, when supplied with effective prompts, attains a consistently elevated occlusion recall across all datasets (approximately 0.82) while sustaining a commensurate precision. This corroborates the framework’s capacity to discern task-relevant regions from textual descriptions alone, the modality-specific anatomical variations notwithstanding (see Figure 3 for visualized exemplars).

Effect of Prompt Type on Region Localization. To further interrogate the influence of linguistic design, we assess the effect that two prompt types exert upon localization performance. As reported in Table 1, descriptive sentence prompts yield superior results on Brain MRI and Chest CT, intimating that richer semantics abet localization. Concise phrases, conversely, prevail on lung radiographs with respect to recall (0.82 versus 0.66), presumably because the broad anatomical purview of the lungs renders general terms sufficiently efficacious.

Effect of SAM Refinement on Region Localizatio. We likewise gauge the contribution of spatial refinement by applying SAM to the coarse masks. As reported in Table 1 SAM refinement augments precision while preserving a high recall. On Brain MRI, for instance, precision ascends from 0.22 to

0.30, even as recall remains above 0.94—a result that attests to the utility of SAM in refining coarse localization outputs.

4.4 COMPARISON WITH PREVIOUS SSL METHODS

We proceed to evaluate the generalization and representational capacity of MWM by juxtaposing it against representative self-supervised learning (SSL) methods drawn from both the reconstruction-based and the contrastive learning paradigms. To ensure an equitable comparison, all methods are pretrained and evaluated under consistent datasets and identical training settings.

Image Classification. Classification performance is scrutinized on three downstream medical imaging datasets. In keeping with common practice, results are reported under two regimes: (1) *full fine-tuning*, wherein the entirety of the model is updated; and (2) *linear probing*, wherein the encoder is held frozen and a solitary linear classifier is trained atop it.

- **Full fine-tuning:** Table 2 collates the classification accuracy of MWM alongside several representative self-supervised methods, namely MAE He et al. (2022), SparK Tian et al. (2023), AnatoMask Li et al. (2024), SimCLR Chen et al. (2020a), MoCoV2 Chen et al. (2020b), and BYOL Grill et al. (2020). MWM uniformly attains the foremost performance, thereby evincing robust generalization across heterogeneous imaging modalities. The most pronounced gain manifests on the Chest CT dataset, where MWM realizes 97.5% accuracy—surpassing SparK by +3.1 and AnatoMask by +1.7. This improvement throws into relief the dividend of text-guided masking, which accentuates semantically critical yet spatially sparse regions—the lung lobes in CT scans, for instance—while forgoing masking redundancy.
- **Linear probing:** To further interrogate the quality of the learned representations, linear probing experiments are conducted. For the sake of concision, MWM is compared against two recent state-of-the-art masked image modeling methods—SparK and AnatoMask. As reported in Table 3 MWM appreciably eclipses both across all datasets, thereby underscoring its aptitude for learning effective features through text-driven pretraining.

Object Detection and Instance Segmentation. To ascertain the generalizability of the learned representations beyond classification, we evaluate brain tumor detection and instance segmentation on the BR35H dataset Hamada (2025). As reported in Table 4, MWM attains the foremost AP^{box} (46.9%) and the second-furthest AP^{mask} (45.9%), thereby surpassing SparK by +1.3 and +1.1 points, respectively. These results attest that MWM transfers efficaciously to both object-level localization and pixel-wise segmentation.

In sum, the incorporation of text-driven masking during pretraining endows the model with more informative and more transferable representations. This semantic guidance yields consistent performance improvements across classification, detection, and segmentation tasks alike, thereby underscoring its efficacy in enhancing visual representation learning.

Method	Type	Brain MRI	Chest CT	Lung X-ray
MoCoV2	CL	95.3	91.7	93.1
BYOL	CL	91.3	93.7	93.9
SimCLR	CL	94.5	93.1	94.1
MAE	MIM	95.4	91.7	94.9
AnatoMask	MIM	96.5	95.8	95.2
SparK	MIM	96.2	94.4	94.7
MWM(Ours)	MIM	96.8	97.5	96.0

Table 2: Fine-tuning classification accuracy (%) on the downstream datasets, wherein MIM designates Masked Image Modeling and CL designates Contrastive Learning.

Method	Brain MRI	Chest CT	Lung X-ray
AnatoMask	70.2	63.9	85.3
SparK	69.8	63.5	81.6
MWM(Ours)	71.3	67.0	88.5

Table 3: Linear probing accuracy (%) on the downstream datasets, evaluated under a frozen encoder.

Method	Detection		Segmentation	
	AP^{box}	AP_{75}^{box}	AP^{mask}	AP_{75}^{mask}
AnatoMask	45.2	53.1	46.2	54.1
SparK	45.6	53.2	44.8	52.0
MWM (Ours)	46.9	53.5	45.9	54.1

Table 4: Detection (AP^{box} , AP_{75}^{box}) and instance segmentation (AP^{mask} , AP_{75}^{mask}) results (%) obtained on the BR35H dataset.

4.5 ABLATION STUDY

We further conduct ablation experiments in order to disentangle the individual bearing of prompt design, SAM-based region refinement, and masking ratio upon the downstream classification performance of MWM.

4.5.1 EFFECT OF PROMPT DESIGN AND REGION REFINEMENT

As reported in Table 5, descriptive sentence prompts outstrip short phrases on brain MRI and chest CT, whereas the converse tendency obtains on lung X-ray. This comports with the localization results of Section 4.3, wherein the prompts that afforded more accurate region identification likewise conducted to superior downstream results. Crucially, both varieties of prompt—irrespective of their granularity—consistently surpass random masking across all datasets. This corroborates the proposition that text-guided masking effects an efficacious semantic alignment between pretraining and downstream tasks, a correspondence that proves pivotal to the enhancement of self-supervised learning performance. Moreover, an ablation upon SAM demonstrates that the incorporation of this module during pretraining consistently improves downstream classification, thereby confirming its efficacy in fortifying text-guided masking.

4.5.2 EFFECT OF MASKING RATIO

As depicted in Figure 4, MWM attains 96.8% classification accuracy at a masking ratio of merely 40%—a setting substantially beneath the roughly 70% conventionally adopted in masked image modeling. Notwithstanding this reduced ratio, it surpasses SparK’s best performance at 70% masking by +2.4% in absolute accuracy. To preclude the possibility that SparK’s inferior performance is an artifact of suboptimal masking configurations, SparK is additionally evaluated under masking ratios of 40%, 50%, 60%, and 70%. The results confirm that 70% masking elicits the highest accuracy for SparK, thereby vindicating the robustness of our comparison. These findings call into question the prevailing supposition that high masking ratios are intrinsically optimal for masked image modeling. They instead accentuate the advantage of semantically guided, task-aware masking strategies such as MWM, which—by conserving task-relevant content while attenuating redundancy—secure stronger downstream performance under lower masking ratios.

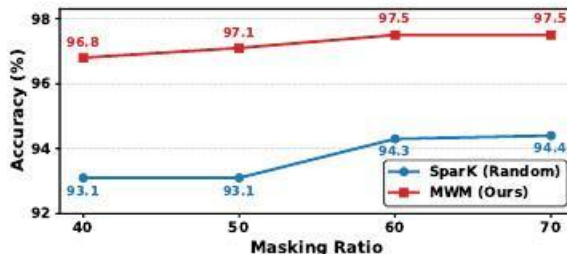


Figure 4: Classification performance on the Chest CT dataset under disparate masking strategies and ratios.

Group	Setting	Brain MRI	Chest CT	Lung X-ray
Prompt Type	No Prompt	96.2	94.4	94.7
	Phrase	96.4	96.5	96.0
	Sentence	96.8	97.5	95.2
Impact of SAM	w/o SAM	96.4	96.8	95.8
	w/ SAM	96.8	97.5	96.0

Table 5: Downstream classification accuracy (%) under disparate prompt types and region refinement strategies across three imaging modalities. *No Prompt* designates random masking bereft of text-guided localization, while *w/* and *w/o* denote the presence and absence of SAM, respectively.

5 CONCLUSION AND LIMITATIONS

This work sets forth **Mask What Matters (MWM)**, a self-supervised pretraining framework that conjoins text-guided semantic localization with region-aware masking. By harnessing natural language prompts to accentuate task-relevant areas and thereupon administering differentiated masking, MWM enables semantically aligned representation learning while dispensing altogether with per-image annotations.

Comprehensive experiments substantiate the efficacy and generalizability of MWM: (1) natural language prompts reliably steer semantic region localization, thereby vindicating the feasibility of text-driven masking; (2) MWM consistently outstrips extant methods (e.g., SparK, AnatoMask) in classification across three imaging modalities, and furnishes concomitant gains in detection and instance segmentation; (3) MWM sustains strong performance even under lower masking ratios (e.g., 40%), thereby underscoring the dividend that semantic guidance confers upon self-supervised pretraining.

Limitations. Promising though it is, our framework is attended by several limitations. First, although text prompts afford flexible guidance, the robustness of MWM to variations in prompt style, vocabulary, or noise remains to be examined systematically. Second, its reliance upon external textual descriptions introduces a dependency liable to enfeeble generalization in settings where dependable prompts are unavailable or inconsistent.

Looking ahead, we contend that text-driven, region-aware masking charts a principled course toward more semantically grounded self-supervised learning, and that the extension of MWM beyond medical imaging may unlock fresh opportunities for broader vision–language pretraining.

REFERENCES

- Sana Aleem, Feiran Wang, Midhun Maniparambil, Eric Arazo, Johannes Dietlmeier, Kevin Curran, Krzysztof J. Geras, Sotirios A. Tsaftaris, Simon J. D. Prince, William M. Wells, William Lotter, Seung Wook Kim, Sebastian Pölsterl, and Samuel Little. Test-time adaptation with salip: A cascade of sam and clip for zero-shot medical image segmentation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 5184–5193, 2024.
- Sourya Basu, Manan Gupta, Chahat Madan, Pulkit Gupta, and Chetan Arora. Focusmae: Gallbladder cancer detection from ultrasound videos with focused masked autoencoders. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 11715–11725, 2024.
- Ting Chen, Simon Kornblith, Mohammad Norouzi, and Geoffrey Hinton. A simple framework for contrastive learning of visual representations. In *Proceedings of the 37th International Conference on Machine Learning (ICML)*, pp. 1597–1607. PMLR, 2020a. URL <https://proceedings.mlr.press/v119/chen20j.html>
- Xinlei Chen, Haoqi Fan, Ross Girshick, and Kaiming He. Improved baselines with momentum contrastive learning. *arXiv preprint arXiv:2003.04297*, 2020b.

- Jin Cheng, Wei Huang, Shuangliang Cao, Riqiang Yang, Wenjia Yang, Zhaoqiang Yun, and et al. Enhanced performance of brain tumor classification via tumor region augmentation and partition. *PLOS ONE*, 10(10):e0140381, 2015.
- Md. E. H. Chowdhury, Tanzila Rahman, Ahsan Khandakar, Rashid Mazhar, Md. Abdul Kadir, Zaid B. Mahbub, and et al. Can AI help in screening viral and COVID-19 pneumonia? *IEEE Access*, 8:132665–132676, 2020.
- Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. Bert: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL-HLT)*, volume 1, pp. 4171–4186, 2019.
- Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, Jakob Uszkoreit, and Neil Houlsby. An image is worth 16x16 words: Transformers for image recognition at scale. *arXiv preprint arXiv:2010.11929*, 2020.
- Ali Fateh, Yalda Rezvani, Sadegh Moayedi, Sam Rezvani, Farzaneh Fateh, and Mohammad Fateh. Brisc: Annotated dataset for brain tumor segmentation and classification with swin-hafnet. *arXiv preprint arXiv:2506.14318*, 2025.
- Fernando Feltrin. Brain tumor mri images 44 classes, 2023.
- Jean-Bastien Grill, Florian Strub, Florent Altché, Corentin Tallec, Pierre H. Richemond, Elena Buchatskaya, and et al. Bootstrap your own latent: A new approach to self-supervised learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 33, pp. 21271–21284, 2020.
- Amit Gupta, Islam Osman, Mohamed S. Shehata, William J. Braun, and Richard E. Feldman. Med-mae: A self-supervised backbone for medical imaging tasks. *Computation*, 13(4):88, 2025.
- Ahmed Hamada. Br35h :: Brain tumor detection 2020, 2025.
- Mohamed Hany. Chest ct-scan images dataset for lung cancer classification, 2020.
- Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pp. 770–778, 2016.
- Kaiming He, Xinlei Chen, Saining Xie, Yanghao Li, Piotr Dollár, and Ross Girshick. Masked autoencoders are scalable vision learners. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 16000–16009, 2022.
- Stanley C. Huang, Abhishek Pareek, Michael Jensen, Matthew P. Lungren, Serena Yeung, and Akshay S. Chaudhari. Self-supervised learning for medical image classification: A systematic review and implementation guidelines. *NPJ Digital Medicine*, 6(1):74, 2023.
- Anukriti Jaiswal, Abhinav Rajendra Babu, Mojtaba Zadeh Zadeh, Debanjan Banerjee, and Fillia Makedon. A survey on contrastive self-supervised learning. *Technologies*, 9(1):2, 2020.
- Daniel Kermany, Kang Zhang, and Michael Goldbaum. Labeled optical coherence tomography (oct) and chest x-ray images for classification. Mendeley Data, v2, 2018. URL <https://doi.org/10.17632/rschjbr9sj.2>
- Alexander Kirillov, Eric Mintun, Nikhila Ravi, Hanzi Mao, Chloe Rolland, Laura Gustafson, and et al. Segment anything. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pp. 4015–4026, 2023.
- Tareq Koleilat, Hamed Asgariandehkordi, Hassan Rivaz, and Yuyin Xiao. Medclip-sam: Bridging text and image towards universal medical image segmentation. In *Proceedings of the International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI)*, pp. 643–653, Cham, 2024. Springer Nature Switzerland.

- Sandor Konya. Ct lung & heart & trachea segmentation, 2020.
- Rahul Krishnan, Pranav Rajpurkar, and Eric J. Topol. Self-supervised learning in medicine and healthcare. *Nature Biomedical Engineering*, 6(12):1346–1352, 2022.
- Yufei Li, Tianyi Luan, Yuhang Wu, Shuhao Pan, Ying Chen, and Xujie Yang. Anatomask: Enhancing medical image segmentation with reconstruction-guided self-masking. In *Proceedings of the European Conference on Computer Vision (ECCV)*, pp. 146–163, Cham, 2024. Springer Nature Switzerland.
- Xin Liu, Fan Zhang, Zhenyong Hou, Lingqiao Mian, Zi Huang, Jian Zhang, and Jinhui Tang. Self-supervised learning: Generative or contrastive. *IEEE Transactions on Knowledge and Data Engineering (TKDE)*, 35(1):857–876, 2021.
- Stuart Lloyd. Least squares quantization in pcm. *IEEE Transactions on Information Theory*, 28(2): 129–137, 1982.
- Bjoern H. Menze, Andras Jakab, Stefan Bauer, Jayashree Kalpathy-Cramer, Keyvan Farahani, Justin Kirby, and et al. The multimodal brain tumor image segmentation benchmark (brats). *IEEE Transactions on Medical Imaging*, 34(10):1993–2024, 2014.
- Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever. Learning transferable visual models from natural language supervision. In *Proceedings of the 38th International Conference on Machine Learning (ICML)*, pp. 8748–8763. PMLR, 2021.
- Tanzila Rahman, Ahsan Khandakar, Yazan Qiblawey, Abdulla Tahir, Serkan Kiranyaz, Sheikh Abul Kashem, Md. Islam, Somaya Al Maadeed, Samar Zughaier, M. A. Khan, and Md. E. H. Chowdhury. Exploring the effect of image enhancement techniques on COVID-19 detection using chest x-rays images. *Computers in Biology and Medicine*, pp. 104319, 2021. doi: 10.1016/j.compbiomed.2021.104319.
- Kai Tian, Yifan Jiang, Qihang Diao, Chaojian Lin, Li Wang, and Zehuan Yuan. Designing BERT for convolutional networks: Sparse and hierarchical masked modeling. In *Proceedings of the International Conference on Learning Representations (ICLR)*, 2023.
- Haoran Wang, Kaiyang Song, Jiashuo Fan, Yibing Wang, Jingdong Xie, and Zhenzhong Zhang. Hard patches mining for masked image modeling. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 10375–10385, 2023a.
- Ying Wang, Tim G. J. Rudner, and Andrew Gordon Wilson. Visual explanations of image-text representations via multi-modal information bottleneck attribution. In *Thirty-seventh Conference on Neural Information Processing Systems*, 2023b. URL <https://openreview.net/forum?id=ECvtxmVP0x>
- Jing Xiao, Yufan Bai, Alan Yuille, and Zongwei Zhou. Delving into masked autoencoders for multi-label thorax disease classification. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, pp. 3588–3600, 2023.
- Shuai Zhang, Yiqiu Xu, Naoya Usuyama, Hongfang Xu, Jayesh Bagga, Rema Tinn, et al. Biomedclip: A multimodal biomedical foundation model pretrained from fifteen million scientific image-text pairs. *arXiv preprint arXiv:2303.00915*, 2023.
- Yiwu Zhong, Jianwei Yang, Pengchuan Zhang, Chunyuan Li, Noel Codella, Liunian Harold Li, Luowei Zhou, Xiyang Dai, Lu Yuan, Yin Li, and Jianfeng Gao. Regionclip: Region-based language-image pretraining. In *2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 16772–16782, 2022. doi: 10.1109/CVPR52688.2022.01629.

A USE OF LARGE LANGUAGE MODELS (LLMs)

We enlisted LLMs *solely* for the purposes of language editing and presentational refinement (e.g., grammar, phrasing, and minor stylistic clarity) of text composed by the authors. LLMs were *not* enlisted for idea generation, experimental design, data annotation, code implementation, or the analysis and interpretation of results. All technical content—algorithms, proofs, figures, tables, metrics, and conclusions alike—was authored and verified by the authors. No confidential or identifying data were furnished to the LLMs beyond anonymized manuscript text.